

BIOL-1 Exam 01 Answer Key

Part A: Multiple Choice (30 points)

1 point each.

Module 01: The Science of Biology & Big Picture

1. **B** (Atom → Molecule → Cell → Tissue → Organism)
2. **B** or **C** (Hypothesis)
3. **C** (Maintain a stable internal environment despite external changes)
4. **B** (The Cell)
5. **C** (Eukarya)

Module 02: Chemistry of Life 6. **A** (Protons) 7. **B** (Covalent bond) 8. **B** (It is less dense than liquid water due to stable hydrogen bonds.) 9. **A** (Acid) 10. **B** (Protons and neutrons)

Module 03: Biological Molecules 11. **C** (Carbohydrates) 12. **C** (Amino acids) 13. **B** (Lipids) 14. **C** (It loses its 3D shape and its function because of heat or extreme pH.) 15. **C** (Nucleic Acids)

Module 04: The Cell & Organelles 16. **D** (It has a membrane-bound nucleus and specialized organelles.) 17. **B** (Mitochondrion) 18. **B** (Ribosomes) 19. **A** (Cell Wall) 20. **B** (Golgi apparatus)

Module 05: The Cell Membrane & Transport 21. **C** (Phospholipid bilayer) 22. **B** (Diffusion) 23. **A** (Water molecules across a semi-permeable membrane) 24. **C** (The cell will lose water and shrink (shrivel).) 25. **D** (Active transport)

Module 06: Energy & Metabolism 26. **D** (Metabolism) 27. **A** (A protein that speeds up chemical reactions.) 28. **D** (ATP) 29. **C** (Chloroplast) 30. **B** (It serves as the final electron acceptor in the electron transport chain.)

Part B: Fill in the Blank (11 points)

1 point each. minor spelling errors are acceptable as long as the intent is clear.

1. **Hypothesis**
 2. **Ionic**
 3. **Active** (site)
 4. **Monosaccharides** (or simple sugars)
 5. **Lysosomes**
 6. **Endosymbiotic**
 7. **Mosaic**
 8. **Denaturation**
 9. **Catabolic** (or exergonic)
 10. **Fermentation**
 11. **Autotroph** (or photoautotroph / producer)
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Part C: Free Response (9 points)

Choose THREE out of five. Each question is worth 3 points. Award partial credit as appropriate for incomplete but partially correct concepts.

1. Water: The Molecule of Life.

- *Key concepts:* Must identify two properties and brief biological examples.
- Example 1: **Cohesion/Adhesion** (allows water transport up plant stems).
- Example 2: **High Specific Heat / Temperature Moderation** (helps organisms maintain stable internal body temperatures; sweating cools body).
- Example 3: **Universal Solvent** (dissolves salts, sugars, and nutrients for transport in blood/sap).
- Example 4: **Density of Ice** (ice floats, insulating bodies of water from freezing solid so aquatic life can survive).

2. The Flow of Genetic Information.

- *Key concepts:* Trace the path through specific organelles.

- **Nucleus** (Manager, where DNA instructions are stored/transcribed).
- **Ribosome** (Factory worker, often on the Rough ER, where the protein is assembled).
- **Golgi apparatus** (Shipping center, modifies and packages the protein into a vesicle for secretion).

3. Crossing the Border.

- *Key concepts:* Differentiate simple diffusion and active transport.
- **Simple Diffusion:** Passive movement of molecules from high concentration to low concentration (down the gradient). Requires **no ATP**.
- **Active Transport:** Movement of molecules from low concentration to high concentration (against the gradient). Requires the input of cellular energy / **ATP**.

4. Energy Cycles.

- *Key concepts:* Connection between photosynthesis and cellular respiration.
- Photosynthesis uses solar energy, CO₂, and H₂O to produce **Glucose and Oxygen**.
- Those products (Glucose and Oxygen) become the reactants for Cellular Respiration.
- Cellular Respiration breaks down the glucose using oxygen to create **ATP**, releasing **CO₂ and H₂O** as waste.
- The waste of one process is the fuel/ingredient for the other, forming a continuous cycle of matter.

5. Molecular Form and Function.

- *Key concepts:* Select a macromolecule, provide an example, and link structure to function.
- **Lipids (e.g., Phospholipids):** Have a hydrophilic head and hydrophobic tails. This specific structure allows them to automatically form a bilayer in water, creating the semi-permeable cell membrane.
- **Proteins (e.g., Enzymes):** Have a highly specific 3D shape containing an active site. This shape perfectly fits a specific substrate, allowing it to catalyze particular chemical reactions.

- **Carbohydrates (e.g., Cellulose or Glycogen):** Cellulose forms long, straight, rigid chains perfect for structural support in plant cell walls. Glycogen is highly branched, allowing for quick enzyme access to release energy rapidly.
 - **Nucleic Acids (e.g., DNA):** The double helix structure with complementary base pairing allows it to be unzipped and easily accurately copied or read for genetic instructions.
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End of exam key.